Day and Night Expected Returns Under Overnight Information Shocks: New Tug-of-War Pattern

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Abstract

This study examines the persistence in overnight and intraday return components in the light of overnight price jumps with a very large dataset of stocks listed on NYSE, AMEX, and NASDAQ. Factoring in overnight information shocks uncovers economically and statistically different *tug-of-war* patterns for day and night return components. When stocks are sorted according to their monthly cumulative overnight returns, a zero-cost portfolio trading strategy yields a 3.9% lower risk-adjusted return for the overnight segment in jump stocks, while the same strategy results in a 4.4% smaller loss during the intraday period in the following month. Findings reveal that the informational strength and surprise component of overnight news prompt adjustments in return patterns across daytime and overnight sections.

JEL classification: G10; G11; G12; G14; G40

Keywords: Price Jumps; Investor Overreaction; Return Predictability; Overnight and Intraday Returns

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1. Introduction

Research on overnight and intraday components of close-to-close returns has revealed new insights for clientele relevance, the content of information, and return predictability.[1] In the asset-pricing context, non-homogeneous investor beliefs and preferences reveal themselves in various forms.[2] Although trading preferences in standard textbook approach rest on representative investor framework, actual market practice is beyond this simplifying assumption, and markets exhibit many deviations from the investor homogeneity premise. One observed heterogeneity is related to market arrival times of different clientele; certain investor groups trade at different parts of the day and that leads to persistent return patterns to the extent these preferences stay perennial. A recent study by Lou et al. (2019) focuses on overnight and intraday return components and ties the observed return patterns to this investor heterogeneity. Specifically, they document return continuation in both overnight and intraday components, along with a reversal mechanism between these segments. This pattern is primarily driven by individual and institutional investors' dominance during different parts of the trading day. In that regard, higher overnight returns in a month are followed by higher overnight and lower intraday returns in subsequent months.

Research conducted thereafter examines this *tug-of-war* between overnight and intraday returns from different perspectives: expanding the discussions further to the intensity of *tug-of-war*, volatility of this intensity, level of investor attention, altered patterns in different Exchanges, and cross-market predictability.[3] A parallel strand of literature demonstrates that investor reactions to news aren't purely mechanical, but are influenced by cognitive biases that affect information processing. Research on overreaction and underreaction to information flows fundamentally stems from these behavioral aspects. The form of information (discrete versus continuous) and heuristic biases play crucial roles in shaping investor reactions and subsequent return patterns in both short

[1] Among others, Jones et al. (1994), Barclay and Hendershott (2003), Jiang et al. (2012), Michaely et al. (2014), Jiang and Zhu (2017), Lou et al. (2019), Akbas et al. (2022), and Barardehi et al. (2022) day and night components of daily returns from those angles.

[2] Seasonality in returns (Ritter and Chopra (1989); Bogousslavsky (2016)), portfolio rebalancing habits (Calvet et al. (2009); Bianchi (2018), trading preferences (Barber and Odean (2008); Berkman et al. (2012); Lou et al. (2019)), consumption and portfolio formation (Bhamra and Uppal (2014)), overreaction and underreaction in returns (De Bondt and Thaler (1985); Jiang and Zhu (2017); Bianchi (2018); Lou et al. (2019); Akbas et al. (2022)) and shocks in market prices (Jiang and Zhu (2017); Frank and Sanati (2018)) are some of the empirical findings linked to this heterogeneity.

[3] Akbas et al. (2022), Cheema et al. (2022), Ham et al. (2022), Bai et al. (2023), Ham et al. (2023), Ma et al. (2023)

and long terms.[4] Despite these contributions, prior literature has not investigated how different informational forms and cognitive aspects affect the *tug-of-war* phenomenon, even though infrequent but unexpectedly large amounts of information impact markets differently than continuous but small information flows, as argued in Da et al. (2014). The effect of overnight information shocks on *tug-of-war* dynamics remains unexplored, and the present study addresses this gap by examining how overnight information shocks shape overnight and intraday return components. Drawing on the psychological principle that dramatic changes stimulate intensified reactions and the tendency to overreact to shocks, this study argues that if discrete information flow triggers distinct return patterns and leads to weakened momentum (as discussed in Da et al. (2014)), then we expect moderated momentum patterns across overnight and intraday segments compared to the findings of Lou et al. (2019) due to aggressive investor reaction to discrete unexpected overnight information.

Revisiting Lou et al. (2019) through the lens of overnight information shocks, this study documents economically and statistically significant changes in return patterns for overnight jump stocks. To put it differently, discrete information arrival leads to significantly different abnormal returns for jump stocks in both overnight and intraday sections for the next month whereas the non-jump stocks exhibit similar trends to those documented in Lou et al. (2019). Succinctly, a zero-cost portfolio trading strategy results in 3.9% less risk-adjusted return in jump stocks for the overnight return component when stocks are sorted according to their monthly cumulative overnight returns although the same strategy ends in 4.4% less risk-adjusted loss for the intraday return component. Additionally, the main *tug-of-war* patterns reported in Lou et al. (2019) are broken for jump stocks in the lowest decile when stocks are sorted according to their overnight return components. Also, the *tug-of-war* phenomenon is intensified for all strategies when stocks are ordered by lagged intraday returns.

Overnight returns and jumps are crucial factors in shaping investor reactions and return dynamics as highlighted in earlier works. For instance, Orlowski et al. (2024) argue that the overnight component of returns exhibits a greater skewness premium, and the associated risk emanates from

[4]For comprehensive discussions on underreaction, overreaction, investor inattention, abundance of news flow, information discreteness and psychological biases, see for instance De Bondt and Thaler (1985), Poterba and Summers (1988), Lo and MacKinlay (1990), Sims (2003), Avramov et al. (2006), DellaVigna and Pollet (2009), Gino and Bazerman (2009), Hirshleifer et al. (2009), Da et al. (2014), and Maćkowiak et al. (2023).

jump and leverage factors. Ham et al. (2022) point out that overnight returns can be instrumental in dissecting investor overreaction due to concentrated news announcement in the overnight section. Overnight returns can also be functional in gauging the level of investor attention. Liu et al. (2023) show that overnight corporate disclosures in Chinese markets attract investor attention and enhance price discovery. Indeed, jumps are critical contributors to the mean return of individual stocks. Kapadia and Zekhnini (2019) demonstrate that stock mean returns mainly accumulate on a few jump days. Jiang and Yao (2013) find that controlling for jumps eliminates the predictive power of size and liquidity factors and substantially reduces the significance of the value factor. The surprising nature of news content behind overnight jumps also aligns findings of this study with the Information Discreteness (ID) discussion of Da et al. (2014) and further extends their momentum analysis from daily return setting to overnight and intraday components.

By integrating overnight information shocks – proxied by overnight jumps- into return predictability discussions, this research contributes to the growing literature on overnight and intraday return dynamics in two aspects. Inspired by Lou et al. (2019), this study first puts forward a new dimension by focusing on informational content and presents that investor sensitivity to overnight information shocks substantially alters *tug-of-war* return patterns across daytime and intraday sections. The adopted approach captures discrete and surprising elements of information that fundamentally shape investor behavior and allows us to uncover different return patterns that would otherwise remain concealed. Hence, by incorporating overnight jumps, this study provides an informational and behavioral framework for *tug-of-war* patterns and demonstrates that the form and intensity of information—not just its presence—play a critical role in shaping return dynamics. As an ex-post metric, overnight jumps encapsulate all forms of information shocks irrespective of news type whereas the firm-specific news in Lou et al. (2019) constitutes only a fraction of aggregate news flow. Moreover, jumps also capture market-wide information shocks from non-idiosyncratic public news, private knowledge, latent information, and other informational events that impact stock groups conjointly. Secondly, this paper aligns jump-driven overnight and intraday return dynamics with investor attention theories and documents an empirical test of Da et al. (2014) for weakened momentum arguments due to overnight information discreteness. Although the ID measure in Da et al. (2014) is built on time-series variability in daily returns, this paper verifies relevant arguments further with the components of daily returns and highlights the impact of discrete information on the interaction of overnight and intraday returns and momentum dynamics with higher data fre-

quency. This dual contribution aims at enhancing our theoretical understanding of price formation and offers insights for market actors that hold overnight positions or investor groups that unload their positions before the end of daytime trading. Zero-cost trading strategies may not directly materialize into profitable strategies due to transaction costs and other frictions that cast a shadow on almost every trading strategy. However, these discussions should be considered, for instance, akin to pinpointing an arbitrage opportunity but not being able to utilize it due to the limits to arbitrage as discussed in Shleifer and Vishny (1997).

The rest of the paper is organized as follows. Section 2 presents relevant literature. Section 3 provides the details of the data and jump detection methodology. Section 4 documents the main empirical findings with intuitive remarks. Section 5 is reserved for the conclusion.

2. Relevant Literature

Previous studies document that overnight returns are mainly shaped by the information accumulated outside daytime trading, and a large extent of firm-specific news is released after market closure. Boudoukh et al. (2019) consider public news as the key factor for overnight returns, and Barclay and Hendershott (2003) document that the very first trades of market opening are the most informative transactions due to public and private information amassed in overnight section. Likewise, Michaely et al. (2014) report that most earnings announcements -81% to 92%- occur outside trading hours. Jiang et al. (2012) similarly analyze S&P500 stocks between 2004-2008 and report that more than 95% of the earnings announcements are made in the overnight section following the market closure. Building on the dominant role of information elements in structuring overnight returns, Jiang and Zhu (2017) use overnight jumps as a proxy for information shocks in their return predictability study. As trading resumes in the morning, other channels of price discovery, such as liquidity, peer signals, and market microstructure, also shape prices alongside information. Trading is halted overnight, and information processing becomes essential in shaping overnight returns. If the content of disclosed information is considerably surprising, investors react aggressively, leading to jumps in the continuous price trajectory. The common theme across this line of literature is that information is the primary driver of overnight returns and a dominant force in opening price formation. Price jumps are particularly consequential within this setting as they shape the mean returns and are instrumental for predictability studies. In that sense, this study unifies with prior

work by treating overnight jumps as essential information events that affect return behaviors.

To date, the relevant literature has improved our knowledge of ex-post and ex-ante returns for overnight and intraday return components.[5] In their recent epochal research, Lou et al. (2019) show that overnight and intraday returns are mainly driven by the interplay between retail and institutional investors, and that creates a persistent pattern in expected returns even into sixty-month horizon. Mostly retail-driven overpricing/underpricing at the market open is typically reversed by opposing clientele trading later in the day. This finding is consistent with Berkman et al. (2012), who report that individual investors -after markets open- snap up stocks that grabbed their attention in the previous day, and with Barber and Odean (2008) who show how higher returns in the preceding day allure retail investors and place them on the buy-side in the next day's opening.

Akbas et al. (2022) analyze *tug-of-war* intensity using the monthly ratio of overnight-intraday return reversal days and scale it by the preceding 12-month average to construct a measure of abnormal frequency. They conjecture that arbitrageurs undervalue the informational content in successive positive overnight returns and attribute these returns falsely to overoptimistic noise trader activity. This results in overcorrection in stock prices. Stocks with frequent 'positive overnight–negative intraday' reversals earn 0.92% higher returns the following month, whereas 'negative overnight–positive intraday' reversals have no predictive power. Inspired by Akbas et al. (2022), Bai et al. (2023) calculate the volatility of daily *tug-of-war* intensity in the Chinese market and report that it significantly predicts monthly stock returns. Authors also document that predictors constructed based on the level and abnormal level of intensity have poor predictive power implying that the market-specific forces in China lead to altered results. In the same vein, Cheema et al. (2022) test the predictive power of *tug-of-war* intensity cross-sectionally in the Chinese market. Contrary to Akbas et al. (2022), future returns do not significantly diverge for stocks with high and low levels of negative daytime reversals. Building on this growing *tug-of-war* literature, this study integrates with the existing literature by showing that overnight jumps systematically alter *tug-of-war* dynamics.

Prior research argues that behavioral and attention-based discussions are crucial: investors do not react uniformly to every instance of information arrival, and substantial overnight price movement affects the timing of transactions by different clientele. Attention is diluted when news

[5] Among others, see for instance Amihud and Mendelson (1987), Ritter and Chopra (1989), Amihud and Mendelson (1991), Gerety and Mulherin (1992), Bogousslavsky (2016) and Jiang and Zhu (2017) for relevant discussions.

is abundant (Michaely et al. (2014), Liu et al. (2023)) and sharpens under discrete shocks (Da et al. (2014), Ham et al. (2023)). This heterogeneous nature of investor response constitutes an essential backdrop for the present study: jumps reinforce attention and generate asymmetric adjustments in return dynamics compared to regular continuous returns. Although Lou et al. (2019) report that opening prices are largely shaped by retail investors, and institutional clients generally trade in the second part of daytime, Ham et al. (2023) analyze an emerging market Exchange and report that all investor types extensively trade with the commencement of trading if the overnight return is surprisingly large in either direction. Liu et al. (2023) highlight that the overflow of firms' overnight announcements may lead to investor inattention and underreaction, confirming the findings of Hirshleifer et al. (2009), which report that earnings announcement days generally coincide, and an overabundance of information on these days leads to diluted investor attention. Likewise, Michaely et al. (2014) highlight that investors' immediate reactions to announcements disclosed outside trading hours is twice that of the earnings surprises announced during trading hours. This is intuitive in the sense that investors have ample time to process the accumulated information in the overnight section, as opposed to trading hours, where the information is digested and priced in less drastically with extended timeframes.

The pattern of information arrival also affects the return dynamics. Da et al. (2014) document that investors devote less attention to continuously arriving small chunks of information, while they exhibit heightened sensitivity to infrequent but influential discrete information flows. Authors show that return momentum is stronger for stocks with continuous information but weaker for those exposed to discrete shocks. Barardehi et al. (2022) conducts this analysis further with overnight and intraday components of daily returns, and their results are consistent with Da et al. (2014).

3. Data and Methodology

3.1. Data

This study uses a raw dataset spanning from July 1993 to December 2021 with stocks from NYSE, AMEX, and NASDAQ available at the CRSP database. The sample includes only common shares, and stocks with multiple PERMNO identifiers for a given PERMCO are excluded. Stocks should also have a continuous trading history without any missing data for trading days. Since jumps are rare events, stocks with a trading history of less than 3 years are not included in the main analyses.

After these filters, slightly less than 9300 stocks remain for the main analysis. As a robustness check, the analyses are repeated by excluding jumps if the accompanying closing price is set to the bid-ask average by the CRSP, and the minimum years of trading history is set to 2 years. During overnight return construction, missing opening prices are set to the previous day's closing price. Daily returns in the CRSP database already reflect the distributions.

The output of the overnight jump filter differs from the data sample of news months in Lou et al. (2019) for several reasons. First, their identification is based mechanically on event calendars and news feeds on Dow Jones Newswire, whereas this study infers the information shocks from ex-post price behavior, where the market revealed its preference and showed what mattered the most for them. Jumps strongly signal unexpected news that is abruptly priced in, unlike Lou et al. (2019), who do not test whether the news was surprising enough to break price continuity. For instance, news months and cases are classified as surprises if the earnings announcements deviate from forecasted numbers in their study, but it does not guarantee any discreteness in information flow nor does not imply that market reaction is powerful enough to create a price jump. Hence, this study focuses on the impact and intensity of overnight information events rather than their mere presence. Overnight jumps are the ex-post measures of surprises regardless of news type and its source. Focusing on realized jumps allows us to pinpoint the situations where market participants actually updated prices in the form of discontinuity. Secondly, the set of informational forms filtered via overnight jumps is considerably larger than the one covered via news months in Lou et al. (2019). One of the most obvious differences is that their study rests only on firm-specific news whereas the overnight jumps additionally filter all forms of public and private information that are not firm-specific and not confined to certain channels. For instance, non-idiosyncratic market-wide news flows may lead to jumps in multiple stocks. It also covers analyst revisions, latent information (e.g., political tensions, geopolitical risks etc), inside trades, litigation news, regulation news related to sectors, rumors, and all other forms of information events that might trigger market unrest or euphoria and lead to excessive price fluctuation enough to create discontinuity. The last difference is related to the technical aspects. Overnight news flows may contain surprising elements resulting in sharp movements at market opening, as well as the elements leading to mild movements with the opening bell. Non-jump stock classification does not imply the absence of information flow; news may still be received, but the information is not shocking enough to alter the price trajectory. As highlighted in Lee and Mykland (2008), being flagged as a jump does not even necessitate

a minimum level of price movement. If the integrated volatility for the relevant jump detection window is low -a characteristic of a calm period-, even small returns may be marked as a jump. On the other hand, disorderly markets inflate integrated volatility, reduce the test statistic, and raise the jump detection thresholds. With a purely objective methodology, the jump detection algorithm analyzes the previous period and produces a numeric output for various forms of news inputs.

3.2. Jump Identification

In the case of the continuous stock price path having discontinuities due to the breaks caused by jumps, one can adopt the following semi-martingale process with a standard diffusive process along with a jump component. Within a given period of [0, T] ($T \geq 0$),

$$dX_t = \mu d_t + \sigma dW_t + J_t \quad ; \quad \forall t \in [0, T] \tag{1}$$

where X is the stock price with drift μ, volatility σ, Wiener process W and jump component J. In the absence of jumps, the price path collapses to the standard Brownian Motion with $dX_t = \mu d_t + \sigma dW_t$. Total price change within the interval [0, T] can accordingly be formulated by

$$X_t - X_0 = \int_0^t \mu d_t + \int_0^t \sigma dW_t + \sum_{i=1}^{N_j^t} J_i \tag{2}$$

where N_j represents the number of jumps for respective price discontinuties of size J_i.

To retrieve the variation due to jumps, Barndorff-Nielsen and Shephard (2004) calculate bi-power variation and subtract it from the total quadratic variation. Let the number of price observations be $K + 1$ within [0, T] with equal time intervals, and Y_{t_i} be the log price at time t_i. One can calculate K returns with the return for the first interval being $r_1 = Y_{t_1} - Y_{t_0}$. The realized variance for [0, T] then becomes $\sum_{i=1}^{K} |r_k|^2$. Encapsulating both the integrated volatility and the variation due to jumps, realized variance converges to quadratic variation as we increase the number of returns with shrinking intervals for a given period [0, T]. As discussed in Barndorff-Nielsen and Shephard (2004),

$$BV = \frac{\pi}{2} \frac{K}{K-1} \sum_{i=2}^{K} \left|r_{k_i}\right| \left|r_{k_{i-1}}\right| \tag{3}$$

$$\lim_{\gamma \to 0} BV = IntV = \int_0^t \sigma^2 dt \quad \text{and} \quad \lim_{\gamma \to 0} RV = QV = \int_0^t \sigma^2 dt + \sum_t \Delta Y_t^2 \tag{4}$$

where γ stands for the size of interval, BV is the bi-power variation, $IntV$ is the integrated volatility, RV is the realized variance and QV is the quadratic variation. This study uses the jump detection technique due to Lee and Mykland (2008) in which the test-statistic $\mathcal{L}_i$ is obtained by standardizing the return realizations with the square root of integrated volatility. That is;

$$\mathcal{L}_i = \frac{r_{m_i}}{\sqrt{IntV_{L\&M}}} \quad \text{where} \quad IntV_{L\&M} = \frac{\pi}{2} \frac{1}{K-2} \sum_{j=i-K+1}^{i-1} \left|r_{m_j}\right| \left|r_{m_{j-1}}\right| \tag{5}$$

in which $IntV_{L\&M}$ stands for the integrated volatility in Lee and Mykland (2008). The return is flagged as a jump if the following inequality is satisfied.[6]

$$\frac{max_{i \in \bar{A}_n} |\mathcal{L}_i| - C_n}{S_n} > \delta \tag{6}$$

This study employs the jump detection algorithm on overnight return time series and stores the days with jumps. As suggested by Lee and Mykland (2008), the length of the window for daily data is set to 16.

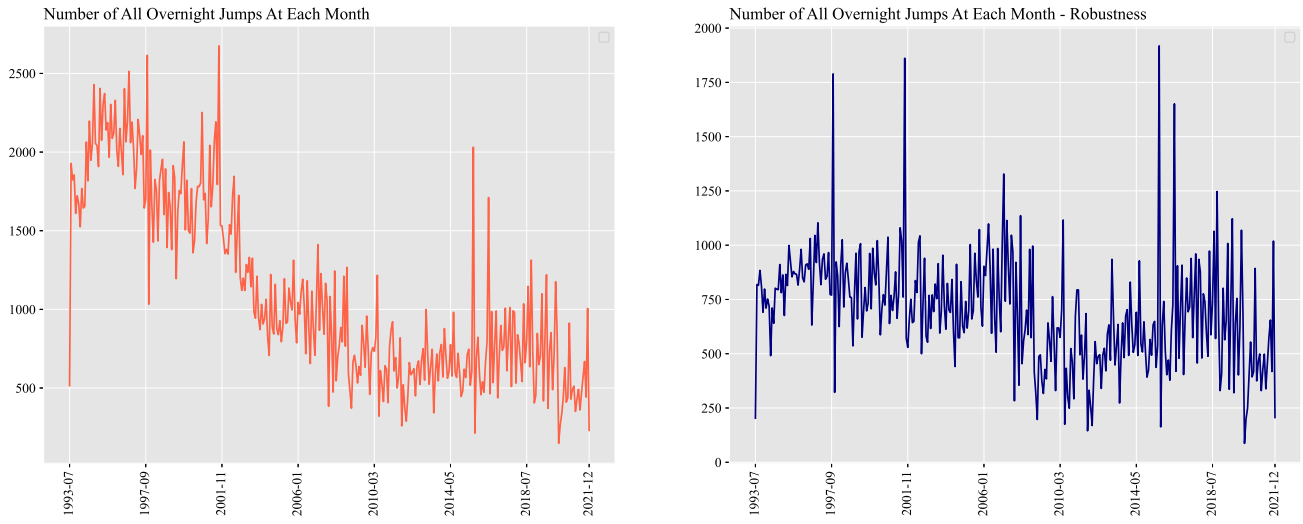
4. Empirical Findings: Overnight Discrete Information and "Tug-of-War"

Within the analysis period spanning from July 1993 to December 2021, 52.8% of total overnight jumps are negative jumps, whereas 47.2% of them are due to positive information shocks. Corresponding figures for robustness check are 49.5% and 50.5% respectively. Figure 1 shows the evolution of monthly jump numbers for the main analysis and robustness setup.

Present study examines how recent information shocks alter overnight and intraday return patterns compared to non-jump stocks. To check that, Table 1 in Lou et al. (2019) is revisited with CAPM and FF4 alphas, and results are reported in Table 1. First of all, stocks are split into overnight jump and non-jump groups each month. They are separately sorted into deciles depending on their cumulative overnight and cumulative intraday return components for this month to implement a

[6] See Lee and Mykland (2008) for the components C_n, S_n, and a comprehensive discussion on jump identification.

Figure 1: Number of Jumps



Notes: This graph shows the number of jumps at each month for the main analysis and robustness check, in which the requirement for minimum trading history is set to 2 years, and jumps are excluded if CRSP sets the closing price to bid-ask average. Since the jump detection algorithm uses the first 16 days as the window for integrated volatility calculation, the number of jumps in the first month stands for the remaining part of the month.

trading strategy that longs the highest decile and shorts the lowest one. Decile portfolio returns are the value-weighted returns every month. For non-jump stocks, results yield quite the same inference as Lou et al. (2019) and confirm the *tug-of-war* pattern. However, according to the findings tabulated in Panel A of Table 1, the overnight portion of the jump stocks in $D10$ produce 1.6% less alpha in the next month, whereas the intraday portion generates 1.8% better relative performance. Along with that, results for the stocks in $D1$ with the lowest overnight returns are also different in jump stocks. Although poor overnight performance persists in non-jump stocks, this is not the case for jump stocks; though insignificant, they have positive risk-adjusted overnight returns of 0.2%. For the intraday returns in the next month, jump stocks in $D1$ have 2.5% less CAPM and FF4 alphas, and their abnormal returns are significant at 10% significance level, although those of non-jump stocks are highly significant with a t-statistic of 4.62. Moreover, long-short trading strategies produce comparably different results for jump stocks. In Panel A, the trading strategy of a long position in $D10$ and a short position in $D1$ in jump stocks produces 3.9% less risk-adjusted return for the overnight section compared to non-jump stocks, whereas the same strategy incurs 4.4% less loss for the intraday portion. As can be seen from Panel A in Table 2, mean differences of jump and non-jump stock portfolios are highly significant for decile portfolios and for the trading

Table 1:
Comparison of Jump and Non-Jump Stocks for 'Tug-of-War'

This table is a retest of Table 1 in Lou et al. (2019) with CAPM and FF4 alphas. The study is performed separately for stocks with and without overnight jumps. At each month, jump and non-jump stocks are determined. Based on their monthly overnight and intraday return components, they are sorted in ascending order, split into deciles, and the overnight and intraday return components are calculated for the next month. Later, the time series of these returns is regressed on the risk factors. Newey-West t-statistic results for 12 lags are reported in parentheses. Panel A and Panel B tabulate results when stocks are ordered according to their overnight and intraday return components, respectively. All the numbers are for the subsequent month. The analyses cover 342 months over the period July 1993 - December 2021.

Panel A: Portfolios sorted by lagged one-month overnight cumulative returns

	Non-Jump Stocks				**Jump Stocks**			
	Overnight		**Intraday**		**Overnight**		**Intraday**	
Decile	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**
D1	-0.022	-0.021	0.036	0.039	0.002	0.002	0.011	0.014
	(-4.12)	(-4.04)	(4.42)	(4.62)	(0.29)	(0.39)	(1.38)	(1.75)
D10	0.046	0.046	-0.041	-0.039	0.030	0.030	-0.021	-0.021
	(6.34)	(6.14)	(-8.14)	(-7.17)	(4.18)	(4.05)	(-5.07)	(-4.49)
D10-D1	0.067	0.067	-0.077	-0.079	0.028	0.028	-0.033	-0.035
	(7.1)	(6.91)	(-7.98)	(-8.05)	(4.89)	(4.54)	(-3.83)	(-4.12)

Panel B: Portfolios sorted by lagged one-month intraday cumulative returns

	Non-Jump Stocks				**Jump Stocks**			
	Overnight		**Intraday**		**Overnight**		**Intraday**	
Decile	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**
D1	0.039	0.039	-0.034	-0.032	0.049	0.049	-0.044	-0.042
	(5.92)	(5.8)	(-7.41)	(-6.57)	(5.89)	(5.74)	(-6.66)	(-5.88)
D10	-0.009	-0.009	0.014	0.015	-0.011	-0.011	0.018	0.019
	(-2.16)	(-2.22)	(2.61)	(2.84)	(-2.18)	(-2.25)	(2.46)	(2.64)
D10-D1	-0.048	-0.048	0.048	0.047	-0.060	-0.060	0.062	0.061
	(-5.89)	(-5.84)	(7.13)	(6.86)	(-6.8)	(-6.66)	(6.52)	(6.21)

strategy when portfolios are formed according to lagged cumulative overnight figures. Altogether, the *tug-of-war* results differ significantly between jump and non-jump stocks.

Panel B reports the results when stocks are sorted according to their cumulative intraday returns. The findings show that all of the results are magnified for overnight jump stocks compared to stocks with no overnight information shock. Jump stocks with the lowest cumulative intraday returns have 1% higher risk-adjusted overnight returns and 1% lower intraday returns compared to non-jump stocks in the next month. For $D10$, jump stocks have 0.2% lower overnight performance and 0.4%

higher abnormal intraday returns. All of the results are statistically significant. The trading strategy of a long position in $D10$ and a short position in $D1$ in jump stocks produces 1.4% more risk-adjusted return for the intraday section compared to non-jump stocks, whereas the same strategy incurs 1.2% more loss for the overnight portion. Again, in Panel B of Table 2, the significance of mean differences are provided when portfolios are formed based on the lagged cumulative intraday returns. Even though means are not statistically different for deciles, the trading strategy raw returns are still statistically different at 10% significance levels.

Table 2:
T-statistic Results for Mean Differences of Jump and Non-jump Stock Decile Portfolios

This table is complementary to Table 1 and tabulates the t-statistic results for mean differences of jump and non-jump stock decile portfolios and trading strategies. If we separately sort jump and non-jump stocks according to their lagged cumulative overnight (intraday) returns and look at the figures in the subsequent month, we will be constructing time series of one-month-ahead return figures for overnight and intraday portions for each decile and trading strategy. This table tells us if the means for jump and non-jump stocks are significantly different from each other in statistical terms. The analyses cover 342 months over the period July 1993 - December 2021.

Panel A: Portfolios sorted by lagged one-month overnight cumulative returns		
	Overnight	**Intraday**
D1	5.60	-3.79
D10	-3.23	3.18
D10-D1	-7.10	7.26
Panel B: Portfolios sorted by lagged one-month intraday cumulative returns		
	Overnight	**Intraday**
D1	1.39	-1.22
D10	-0.84	0.93
D10-D1	-1.72	1.88

On average, stocks with overnight negative (positive) jumps have positive (negative) intraday cumulative returns. These shocks intensify return reversals relative to the reversal figures in tranquil days, consistent with the findings in Panel B of Table 1. In Panel A, results of $D1$ are not significant and do not conform to the basic *tug-of-war* pattern. Even though they have a positive intraday risk-adjusted return of 1.4% (significant at 10% confidence level), the negative performance in the overnight section does not extend to next month. Reported *tug-of-war* findings are also consistent with Da et al. (2014), in which the authors show that momentum returns are more pronounced with continuous information stocks than stocks with discrete information. They report that investors are less attentive to frequently arriving small chunks of information and more heedful and alert to the intermittent but attention-grabbing news flows. To put it differently, small but continuously arriving information does not catch investors' attention, and unprocessed information creates deviation from

Table 3:
Comparison of Jump and Non-Jump Stocks for 'Tug-of-War'
Robustness Check: At least two years of trading history & Overnight jumps eliminated if CRSP closing price is set to bid-ask average

This table is a retest of Table 1 in Lou et al. (2019) with CAPM and FF4 alphas. The study is performed separately for stocks with and without overnight jumps. At each month, jump and non-jump stocks are determined. Based on their monthly overnight and intraday return components, they are sorted in ascending order, split into deciles, and the overnight and intraday return components are calculated for the next month. Later, the time series of these returns is regressed on the risk factors. Newey-West t-statistic results for 12 lags are reported in parentheses. Panel A and Panel B tabulate results when stocks are ordered according to their overnight and intraday return components, respectively. All the numbers are for the subsequent month. The analyses cover 342 months over the period July 1993 - December 2021.

Panel A: Portfolios sorted by lagged one-month overnight cumulative returns

	Non-Jump Stocks				**Jump Stocks**			
	Overnight		**Intraday**		**Overnight**		**Intraday**	
Decile	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**
D1	-0.023	-0.022	0.038	0.041	0.005	0.005	0.009	0.011
	(-3.99)	(-3.91)	(4.31)	(4.47)	(0.95)	(1.07)	(1.23)	(1.63)
D10	0.043	0.043	-0.039	-0.037	0.030	0.031	-0.023	-0.022
	(5.77)	(5.6)	(-7.88)	(-7.14)	(4.13)	(4.01)	(-5.7)	(-4.87)
D10-D1	0.065	0.065	-0.077	-0.078	0.025	0.025	-0.032	-0.033
	(6.81)	(6.64)	(-7.48)	(-7.55)	(4.66)	(4.38)	(-4.32)	(-4.53)

Panel B: Portfolios sorted by lagged one-month intraday cumulative returns

	Non-Jump Stocks				**Jump Stocks**			
	Overnight		**Intraday**		**Overnight**		**Intraday**	
Decile	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**	**CAPM**	**FF4 alpha**
D1	0.037	0.037	-0.035	-0.033	0.050	0.050	-0.047	-0.044
	(5.5)	(5.38)	(-7.54)	(-6.51)	(5.71)	(5.54)	(-7.17)	(-6.35)
D10	-0.009	-0.009	0.014	0.015	-0.010	-0.010	0.016	0.017
	(-2.21)	(-2.26)	(2.59)	(2.83)	(-2.33)	(-2.44)	(2.55)	(2.82)
D10-D1	-0.046	-0.046	0.048	0.048	-0.060	-0.060	0.063	0.062
	(-5.62)	(-5.57)	(7.3)	(6.98)	(-6.53)	(-6.36)	(6.67)	(6.31)

rational prices, creating abnormal returns for various trading strategies. With information shocks, inattention fades and investors become more vigorous, leading to derailed prices being routed back toward their rational path.

ID in Da et al. (2014) comes along with low persistence and elevated correlation during short window events. As highlighted in Barardehi et al. (2022), daytime private information is gradually absorbed, and delayed investor learning results in momentum, and the incorporation of private

information into price requires voluminous trading. Da et al. (2014) and DellaVigna and Pollet (2009), among others, highlight that investors pay little attention to factors showing low periodic fluctuations. Discrete information that is considered a market jolt, on the other hand, leads to concerted market reaction. In other words, heightened attention results in spiked correlation, and investors act in tandem. Buccheri et al. (2021) for instance report that conditional correlations, on a granular level, jump at 14:00 pm in response to Federal Open Market Committee meetings. In the same vein, if the aggregated overnight information contains a surprise component, investors react aggressively, leading to weakened momentum. Da et al. (2014) also associate the estimated positive Fama-Macbeth regression coefficient of ID as a sign of a premium for jump risk, and the interaction of ID with PRET (the previous returns) has negative loadings in all model specifications; an indication of a downward adjustment in momentum returns. Moreover, Da et al. (2014) report that ID is not persistent with a very low autocorrelation level of 0.033, and that is compatible with the nature of jumps which are inherently rare. Likewise, Boyer and Vorkink (2014) point out that low probability events are not persistent and lagged skewness is not instrumental for ex-ante skewness. Hence, results that lean on price jumps, which are sporadic and independent in a general sense, may also be considered from the same perspective. The conjecture that overnight jumps weaken the momentum dynamics and leads to altered *tug-of-war* patterns is grounded on discrete information resulting in collective action and spiked attention. Jumps' rare nature similarly conforms to the low levels of persistence.

Findings displayed in Panel A of Table 1 conform to the intuitions in Da et al. (2014): FF4 alphas are smaller in magnitude in jump stocks compared to non-jump stocks. Portfolios formed according to lagged one-month overnight cumulative returns have smaller alphas in overnight jump stocks in Panel A. On the other hand, all of the magnitudes in Panel B are higher in jump stocks when portfolios are formed according to lagged one-month intraday cumulative returns. Since these are overnight jump stocks as opposed to stocks with intraday discontinuities, the results are naturally altered. Overnight jumps proxy information shocks in the wake of market closures and which fosters the discreteness of information in those stocks and debilitates the return continuation and momentum. In Panel A - Table 1, negative and positive overnight returns ($D1$ and $D10$) continue into the next month for non-jump stocks; negative (positive) overnight returns are followed by negative (positive) overnight returns as the information is less discrete. However, the discreteness of the information in jump stocks impedes the continuation of overnight negative returns into the

next month; though it lacks statistical significance, next month's overnight returns become positive (at least not negative anymore). In the same fashion, jump stocks with positive overnight returns ($D10$) in one month have shrunk four-factor alphas in the next month. Since the sorting scheme in Panel B is one-month cumulative intraday returns, although the discreteness of information emanates from overnight news shocks, we see amplified alphas for jump stocks in all the intraday sections. As a consequence of the inflated monthly intraday returns driven by the overreaction to discrete overnight information arrivals, jump stocks have escalated returns in the intraday section for both extreme decile portfolios and the zero-cost portfolio.

Table 3 documents 'tug-of-war' results with stocks with at least 2 years of trading history. If the CRSP closing price on jump day is set to the bid-ask average, these jumps are also excluded from the analyses. The results are almost the same as the base scenario; abnormal returns are not altered except for immaterial differences. Significance levels are also very close to the ones in Table 1.

5. Conclusion

This study shows that isolating jump stocks (stocks with discontinuities in their prices) and non-jump stocks can uncover different return dynamics that would otherwise have been concealed due to hoarding. To exemplify how this separation substantially alters results and uncovers new return dynamics, the *tug-of-war* phenomenon -which was recently documented by Lou et al. (2019)- is revisited. Focusing on overnight information shocks resulted in statistically and economically different expected returns for overnight and intraday components for the next month. When stocks are sorted according to their intraday return components, the *tug-of-war* pattern is amplified. When sorting is by overnight components, however, *tug-of-war* findings become considerably less in magnitude. These findings suggest that the strength and nature of information play a significant role in shaping overnight and intraday return dynamics. The prior literature also documents that return patterns are affected by both market-level and stock-specific ambiguity. Examining the interplay between these return components in periods of elevated ambiguity remains a promising avenue for future research.

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